PRELIMINARY DESIGN – FOR MDR INTEGRATION

Raspberry Pi Pico Electronics Architecture

Process flow, component rationale and 3K design-criteria fit

RTC 2027 Student 3K | Mission Design Report: electronics contribution | 4 November 2026

Scope and status. This is the electronics team's contribution draft to the team's single MDR; it is not a separately submitted report and not a completed flight design. Competition constraints are attributed to the organiser documents [1], [2], [3]. Thresholds, rates, pin allocations, power figures and the thermal payload are *team proposals* marked as such, and require simulation, approval and later testing in line with the MDR/PDR/CDR maturity expectations [3], [4]. Manufacturer figures are quoted from the cited datasheets and should be re-checked against the exact module purchased.

1. Technical issue closure checklist

The review of the previous draft raised six technical issues. This revision addresses each one as follows. "Status" states honestly what is closed in the document and what still needs team input or hardware evidence.

ID	Technical issue	Closure action in this revision	Status	Where
☐ A	Three Picos are not automatically independent recovery channels.	Common-cause analysis; diverse barometer and algorithm in channel B; release-interface requirements R1–R4; fault-injection tests.	Analysis done; release mechanism still TBD (recovery lead).	Sec. 8
☐ B	MPU6050 ±16 g range is not justified against the flight.	Specific-force equation, input list and a numerical decision gate for adding the ±200 g ADXL375; saturation flag.	Rule defined; needs Pioneer-10K thrust curve and simulation TBD.	Sec. 9
☐ C	Watchdog scratch registers do not protect against power loss.	Corrected claim; reset-cause table; FRAM checkpoint with CRC and double slots; hold-up capacitance; cold-boot rules; revised Fig. 2.	Design defined; to be demonstrated by V-04.	Sec. 10
☐ D	Thermal payload lacks a defined scientific objective.	Research question, hypothesis, uncertainty budget, completeness and retrieval criteria, and what new information results.	Defined; accuracy to be demonstrated by V-08.	Sec. 11
☐ E	Budget omits recovery controllers, protection, mounting and verification.	Tiered budget separating Shopee-priced parts, allowances and unquoted items, with spares, test costs, contingency and mass estimate.	Provisional; release hardware and GPS beacon unquoted.	Sec. 14
☐ F	GPS and flight-data outputs lack a defined interface.	Post-flight data pipeline, time-alignment method, merged table and per-output availability criteria.	Defined; beacon file retrieval method to confirm with organiser.	Sec. 15

2. Design drivers: how the Pico design meets the 3K criteria

The design uses **three separate Raspberry Pi Pico boards** (RP2040) [5], [6]: one *logger/payload* Pico that never commands recovery, and two *independent recovery* Picos (primary and backup), each with its own barometer and battery. This keeps required flight-data logging and the thermal experiment from sharing a failure path with parachute deployment, while using one familiar, low-cost, well-documented platform for every computer in the rocket.

3K criterion	Design response	Basis / status
Record altitude, velocity, acceleration, GPS trajectory	Logger Pico records BMP280 pressure-altitude and MPU6050 acceleration with timestamps; velocity is derived by filtering/fusion; GPS track is taken from the organiser beacon and merged post-flight (Sec. 15).	[1] Sec. 2.1; proposed method
Detect launch, apogee, descent	Multi-condition state machine (Figs. 2–3) with lock-outs and time-out back-ups.	[1] Sec. 4.2.2.4; proposed
Two-stage recovery: drogue at apogee, main at lower altitude	Two recovery Picos each execute both events; main altitude selected in the 150–450 m window, below the 500 m inspection criterion.	[1] 4.2.2.4; [7] Row 8

3K criterion	Design response (continued)	Basis / status
Independent / redundant activation, release redundancy	Primary and backup recovery Picos, separate batteries, separate barometers, diverse barometer and algorithm in B, outputs held low in hardware (Fig. 5); common-cause analysis in Sec. 8.	[1] 4.1.3, 4.2.2.2; mechanism TBD
Safe/armed states, arming confirmation	Arming switch gates power; outputs forced low by pull-downs; launch-inhibit logic; buzzer/LED status code.	[1] 4.6.3–4.6.7
Vibration, impact, sensor error	Plausibility gates, filtering, CRC records, strain-relieved connectors, watchdog restart and FRAM state checkpoint (Sec. 10).	[1] 4.1.2, 4.4
> 1.5 h electrical operation	Single Li-ion cell per Pico; load ≈56 mA estimated, 200 mA design basis (Sec. 13).	[1] 4.7.2; [7] Row 6
Official GPS tracker	Separate, organiser-standard beacon, own power, no electrical dependence on Pico boards.	[1] 2.2, 5.1
Additional payload	Two DS18B20 probes read by the logger Pico: internal avionics-bay vs inner-wall temperature difference with defined research question (Sec. 11); retained in rocket, no radio, no stored-energy device.	[1] 7.1; team proposal
No unapproved radio emissions	Plain Pico (not Pico W): no on-board radio.	[1] 4.1.4; [5]

3. System architecture

Critical interface rule. The logger never commands recovery, and nothing on Domain L (SD card, thermal probes, buzzer) can block a recovery decision. Cross-links between domains are limited to a common ground reference at the release interface if the recovery team requires it; no shared supply [1].

4. Why the Raspberry Pi Pico

RP2040 / Pico feature	What it provides	Benefit to the 3K rocket
Dual Cortex-M0+ cores, up to 133 MHz	Two independent cores sharing memory [6].	Core 0 senses and decides; core 1 does slow SD writes, so storage latency cannot delay sampling (Fig. 4).
264 kB on-chip SRAM	Fast RAM for buffers [6].	A 5 s pre-trigger and write-ahead buffer at 24 B × 200 Hz uses about 24 kB (≈9% of SRAM).
Hardware I^2C, SPI, UART, ADC	Two I^2C, two SPI, two UART, 12-bit ADC inputs [6].	Covers IMU/baro (I^2C), SD (SPI) and battery-voltage monitoring without extra chips.
Programmable I/O (PIO)	Eight state machines for custom protocols [6].	Cycle-exact 1-Wire for DS18B20 without blocking the CPU.
Watchdog with scratch registers	Reset timer; scratch registers survive a watchdog reset but not a power-on reset [6].	A watchdog reset recovers flight state (Fig. 2); genuine power loss needs separate checkpointing (Sec. 10).
Wide supply: VSYS 1.8–5.5 V	On-board buck-boost supply to 3.3 V [5].	A single 3.0–4.2 V Li-ion cell powers the board directly; the rail holds as the cell discharges.
No radio (Pico, not Pico W)	Plain board without Wi-Fi/Bluetooth [5].	Avoids unapproved emissions from the rocket [1].
Open C/C++ SDK, SWD debug, UF2 boot	Free toolchain and bootloader [8].	Rapid iteration, scripted builds, and the same toolchain on all three boards.
Low cost and availability	Retail boards at a few ringgit, drop-in spares.	Three flight computers plus spares fit a student budget; identical hardware simplifies sparing.

Alternatives considered. An 8-bit AVR such as the ATmega328P has only 2 kB of SRAM [9], too little for a buffered logger plus filtering. ESP32 devices offer more processing but include Wi-Fi/Bluetooth radios [10] that must be disabled or justified under the radio rules [1]. The Pico gives enough memory and I/O without a radio at similar cost.

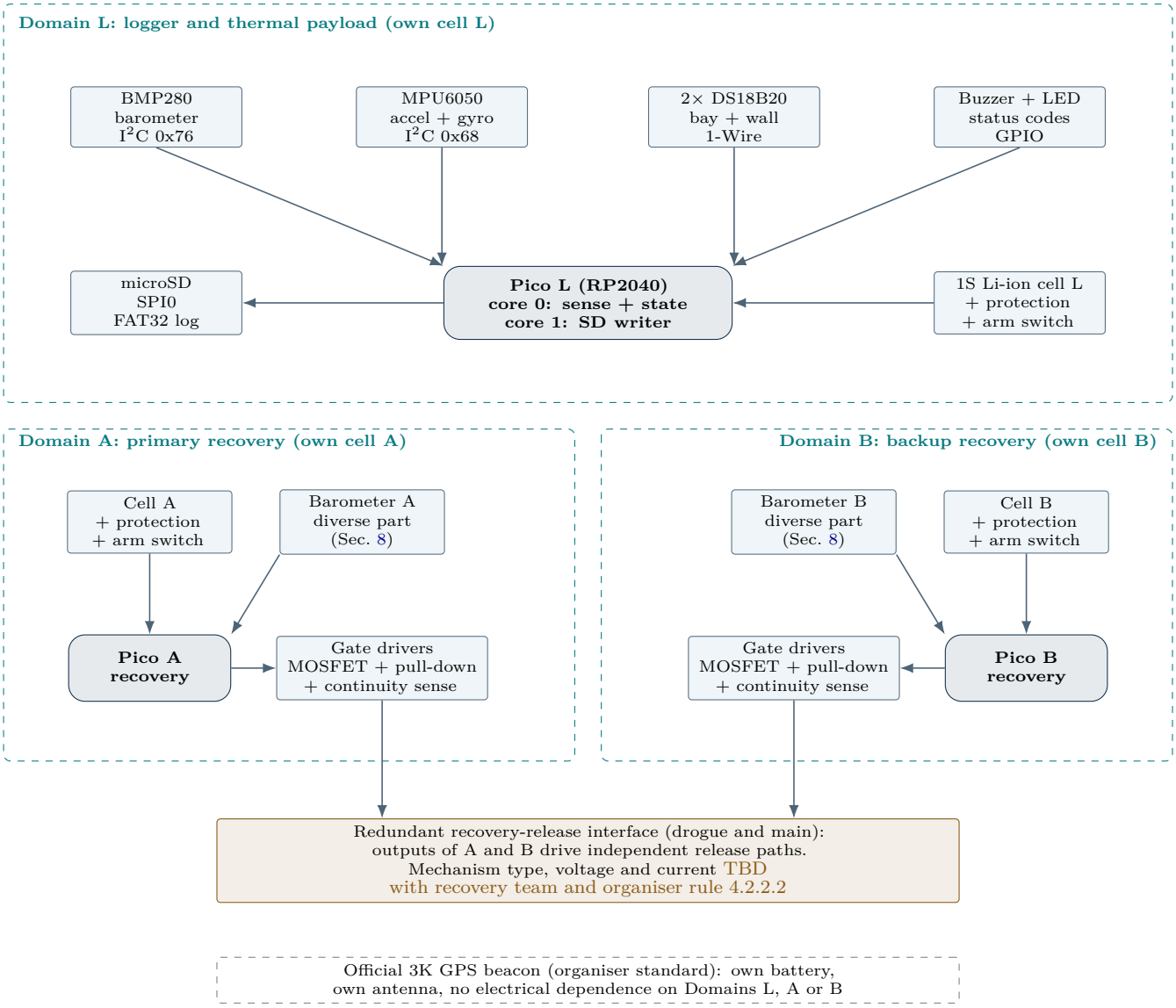


Figure 1: Electronics architecture. Three Pico boards in three power domains; the logger can fail without affecting recovery, and recovery A can fail without affecting recovery B. A FRAM checkpoint memory on each Pico (I^2C1, Sec. 10) is omitted for clarity. Illustrative; wiring and fault independence remain to be approved.

Known Pico limits and mitigations. The Cortex-M0+ has no hardware FPU [6]; at 200 Hz the filter workload is small, but fixed-point or measured float timing will be checked. Flash is external (XIP), so *no flash writes during flight*; recovery events are kept in RAM and dumped after landing [6], [8]. A Pico dev board is not qualified flight hardware: the independence of A and B, the watchdog, hard-wired pull-downs and testing are what give confidence, not the board itself. The Pico operating range is −20 to +85 °C [5]; the bay temperature is measured by the payload.

5. Component-by-component rationale

Component	Role	Rationale and fit to 3K criteria	Limit / mitigation
Pico L (logger)	Flight data, state estimate, thermal payload, status.	Satisfies mandatory output list [1]; isolates non-safety functions from recovery.	Not a recovery path. Verified by Fig. 1 separation.
Pico A / Pico B (recovery)	Independent apogee and main-altitude logic and output control.	Redundant activation and release redundancy [1]; identical code base reduces effort, separate boards and cells remove single points of failure.	Common-cause risk reduced by diversity, not eliminated (Sec. 8).

Component	Role	Rationale (continued)	Limit / mitigation
BMP280 (logger, recovery A)	Pressure/temperature → altitude.	Pressure range 300–1100 hPa covers sea level to well above 3 km (≈701 hPa at 3 km in a standard atmosphere); relative accuracy about ±0.12 hPa (≈ ±1 m) [11]. Barometric altitude is the standard method for apogee and main-deployment decisions.	Port must see static pressure, vented away from fast airflow; lock-out near burnout/transonic transients; absolute offset removed by pad baseline.
MPU6050	Acceleration (and angular rate) for launch/burnout detection and acceleration record.	16-bit ADC, programmable ±2 to ±16 g, 400 kHz I^2C [12], [13]; ±16 g gives 2048 LSB/g, ample resolution.	Range decision gate for the ±200 g ADXL375 [14] in Sec. 9; saturation flagged; gyro ±2000°/s limits roll-rate capture.
DS18B20 ×2	Payload temperatures: avionics bay and inner airframe wall.	Digital 1-Wire, factory calibrated, −55 to +125 °C, ±0.5 °C (from −10 to +85 °C), selectable resolution up to 12-bit [15]; many sensors share one wire; 1 Hz sampling matches the conversion time.	Probe thermal lag; mounting bias; cable strain relief. Powered normally (not parasitic).
microSD + SPI module	Non-volatile flight log.	SD cards support SPI mode [16] and are handled by the open FatFs library [17]; ≈26 MB for 90 min at 4.8 kB/s, so even small cards are oversized.	Use a 3.3 V-native socket and a ≤32 GB FAT32 card; log with 512 B blocks and 1 s syncs; test under vibration.
1S Li-ion cell per board (A, B, L)	Power.	Compact, high energy; 3.0–4.2 V maps to the Pico VSYS window [5]; separate cells keep faults isolated (battery safety per [1] 4.7).	Needs protection circuit, fuse, secure holder; verify endurance > 1.5 h.
MS5611 barometer (recovery B)	Diverse second barometer for channel B.	A different sensor part avoids a shared systematic fault or supply-chain defect in a single part number [18], [19].	Different driver and calibration maths; allowance in budget, price unquoted.
I^2C FRAM (each Pico)	Non-volatile state checkpoint.	Survives power loss, writes in well under 1 ms and is not subject to the flash-write restriction in flight [6], [20].	Separate bus (I^2C1); CRC and double-slot writes (Sec. 10).
Arming switch + pull-downs	Safe/armed states.	Hard power gate and gate pull-downs keep outputs low regardless of software state, supporting the safe/armed rules [1].	Accessible from outside the rocket; status code on buzzer/LED.
MOSFET gate drivers + continuity sense	Switch the release device, check the circuit.	Logic-level MOSFETs let 3.3 V GPIO switch higher currents; low-current continuity check proves wiring without activating the release.	Device current and voltage TBD; flyback/EMI protection required.
Buzzer + LED	Self-test and locator.	Immediate pad confirmation of normal operation/arming status [1]; post-landing locator aid.	Not relied on for any flight decision.
JST-XH, labelled harness, PCB/protoboard	Interconnect.	Locking connectors, strain relief and labelled wiring support the vibration and wiring rules [1].	Retention and pull tests.
Official GPS beacon	Location after landing.	Required organiser equipment; kept electrically separate [1].	Installation and cost to confirm.

6. Interfaces and pin allocation (proposed)

Pico	Signal group	Proposed allocation and notes
Logger (L)	I^2C0 (GP4 SDA, GP5 SCL), 400 kHz	BMP280 and MPU6050 on one bus; 4.7 kΩ pull-ups (breakouts may already add them). Keep leads short [13].
Logger (L)	SPI0 (GP16–GP19)	microSD, 3.3 V logic; CS pull-up.
Logger (L)	1-Wire via PIO (GP15)	4.7 kΩ pull-up; both DS18B20 on one bus, addressed by ROM code, conversion started for both together.
Logger (L)	ADC0 (GP26)	Cell voltage via resistor divider.
Logger (L)	GP14 buzzer, GP25 LED	Status codes only.
Recovery A/B	I^2C0 (GP4/GP5)	Own barometer per board (A: BMP280, B: MS5611 [18]); not shared with L.
Recovery A/B	GP10 drogue, GP11 main	To MOSFET gates; external 100 kΩ pull-down on each gate in addition to internal pulls.
Recovery A/B	GP12/GP13, ADC0	Continuity sense inputs and cell voltage.
All Picos	I^2C1 (GP6 SDA, GP7 SCL)	FRAM checkpoint memory on a bus separate from the sensors, so a sensor-bus fault cannot block state saving [20].

7. Process flowcharts

Figures 2–5 describe the software processes. Rounded boxes are start/end, rectangles are processes, diamonds are decisions, and amber boxes are fault or degraded paths. Parameters are in Sec. 7.1.

7.1 Decision parameters (initial placeholders pending simulation)

Symbol	Meaning	Starting value / source
a_L, t_L	Launch acceleration threshold and duration (logger and recovery).	e.g. 2.5 g for 50 ms (10 IMU samples); set from motor thrust-to-weight in simulation.
h_L	Altitude rise confirming launch.	e.g. 10 m above baseline within 2 s.
t_{lock}	Apogee lock-out after burnout.	Simulation-derived, e.g. 1–2 s, to avoid pressure disturbances near burnout/transonic flight.
N	Consecutive filtered samples with $\dot{h}_f \leq 0$.	5 samples at 25 Hz (0.2 s).
$T_{apo,max}$	Apogee time-out from launch.	Simulated apogee time + margin.
h_d (A / B)	Drop below peak that confirms apogee on recovery boards.	A: small (e.g. 3 m); B: larger (e.g. 8 m) plus delay Δt_B.
h_{main} (A / B)	Main-parachute deployment altitude (AGL).	Within 150–450 m [1], [7]; B set about 30 m lower than A.
t_{pulse}	Output pulse duration.	From release-device specification TBD.
v_{max}	Baro plausibility limit on $\lvert dh/dt \rvert$.	Simulated peak speed × margin TBD.

8. Recovery independence and common-cause failure analysis

Separate boards, cells and barometers remove many single-point failures, but they do not by themselves make two channels independent: a shared design defect, a shared pressure environment or a shared release element can defeat both. Software "independence" in particular should not be assumed, because independently written programs have been shown to fail on the same inputs more often than independence would predict [19]. The design therefore combines *diversity*, a *qualitative common-cause analysis* and *fault-injection tests*. Redundancy reduces common-cause risk; it does not eliminate it.

Shared element	Common-cause failure	Countermeasure	Residual / evidence
Software logic	Same defect or same wrong threshold in A and B.	Separate firmware builds written and reviewed by different members; B adds a time-since-launch back-up and different h_d and h_{main} (Sec. 7.1).	Diversity helps but is imperfect [19]; V-01 replay, V-05.
Barometer part	Systematic offset, noise or supply defect in one part number.	A uses BMP280 [11], B uses MS5611 [18]; different drivers and calibration code.	Both read the same atmosphere; see port row.

Shared element	Common-cause failure	Countermeasure (continued)	Residual / evidence
Static pressure port	Blocked or disturbed port, bay leak, transonic or burnout pressure transient.	Separate ports at different circumferential positions; lock-out t_{lock}; B back-up timer; vented-bay pressure test.	Residual common pressure field; V-09.
Power	Single cell or connector failure.	Separate cells, switches and harnesses; no shared supply; common ground only at the release interface if required.	V-06, V-08.
Release mechanism	One mechanical or electrical element defeats both channels.	Requirements R1–R4 below; independent initiators per channel.	Mechanism TBD; V-05.
Environment	Vibration, shock, temperature or EMI affects both boards.	Separate mounting, distance between boards, strain-relieved wiring, measured bay temperature.	Vibration and thermal tests.
Human error	Wrong thresholds, wiring or arming sequence on both boards.	Configuration checksum shown on the status code at arming, labelled harnesses, two-person checklist.	Inspection at integration.

Release-interface requirements (proposed).

R1. Each channel (A and B) shall be able to release both drogue and main without the other channel's power or logic.

R2. No single mechanical or electrical element shall prevent both channels from releasing, consistent with the release-redundancy rule [1] (Sec. 4.2.2.2); if a mechanism is shared, the shared element and its redundant initiators shall be documented.

R3. All outputs are de-energised by default, and each channel's supply passes through its own arming switch.

R4. Compliance is shown by fault injection: A disabled, B disabled, and both enabled, each with safe test loads (V-05).

Candidate arrangements, to be selected with the recovery lead: (i) a common release mechanism with two independent initiators, one per channel; or (ii) two separate mechanisms, subject to the rules. The choice is needed before the MDR text is frozen so that this interface is no longer TBD.

9. Accelerometer range decision

The MPU6050 offers programmable ranges up to ±16 g [12]. Whether that is enough depends on the rocket. An accelerometer measures specific force along its axis, which in g is

$$a_{sf}(t) = \frac{T(t) - D(t)}{m(t)\, g_0}, \tag{1}$$

where T is thrust, D drag, m the (decreasing) mass and $g_0 = 9.81\,\mathrm{m/s^2}$. Parachute-deployment shock must also be included. The inputs are listed below; the Pioneer-10K thrust curve and the team flight simulation were not part of the material used for this revision, so the numerical result is TBD.

Input	Use	Status
Pioneer-10K thrust curve $T(t)$ and burn time	Peak thrust and thrust tolerance.	TBD
Mass history $m(t)$, burnout mass	Peak a_{sf} occurs near low mass.	TBD
Drag $D(t)$ from flight simulation	Net specific force, transonic region.	TBD
Drogue and main deployment shock	Short-duration peaks that may exceed boost peak.	TBD (analysis, later test)
Margin factor k	Covers thrust variation and simulation error.	$k = 1.25$ (team assumption)

Decision rule. Let a_{pk} be the largest predicted $|a_{sf}|$ over the whole flight including deployment shock. If $k\,a_{pk} \le 16$ g, the MPU6050 alone is adequate for the acceleration record. If $k\,a_{pk} > 16$ g, a ±200 g ADXL375 [14] is added to the logger as a second acceleration channel (budget line in Sec. 14). In every case the logger sets a saturation flag when a raw sample reaches full scale, and launch detection still works because it is a threshold crossing well below full scale. Barometric velocity and altitude remain available even if the acceleration record clips. Verification is analysis V-07.

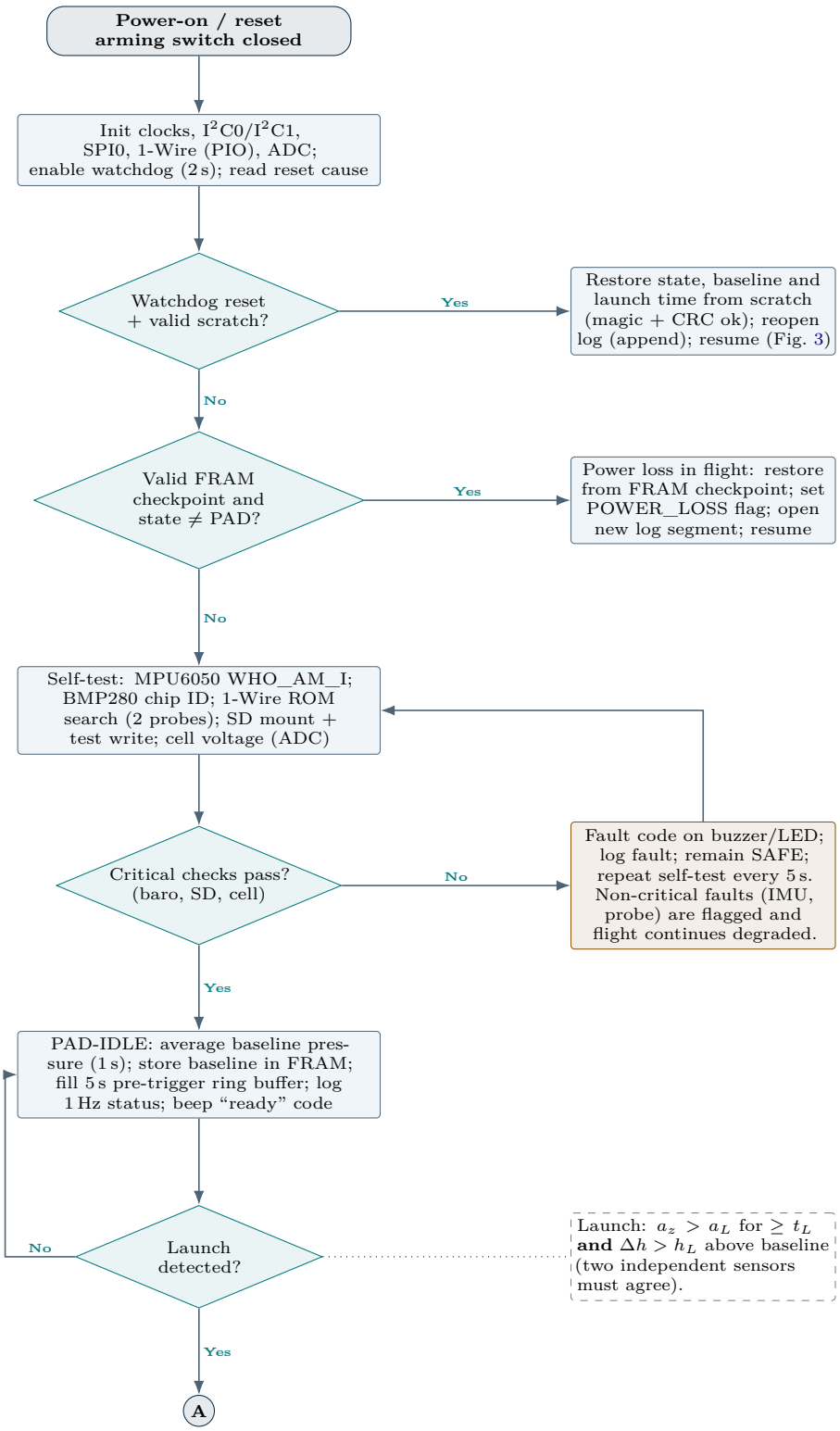


Figure 2: Logger Pico start-up with reset-cause discrimination: a watchdog reset restores state from scratch registers; a power loss restores it from the FRAM checkpoint; otherwise a normal self-test and pad-idle run until launch detect.

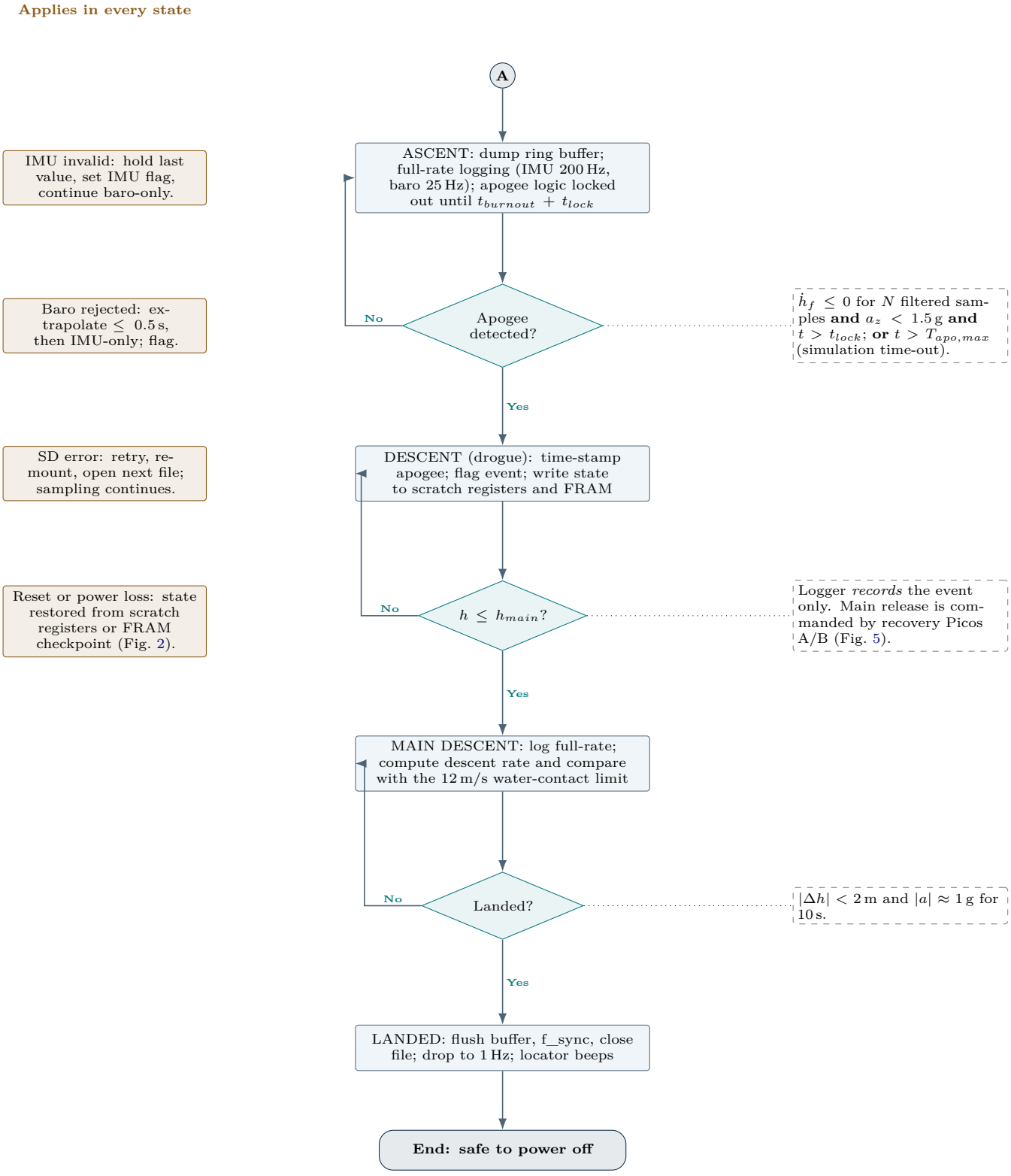


Figure 3: Logger Pico in-flight state machine. Left column: anomaly handling that applies in all states.

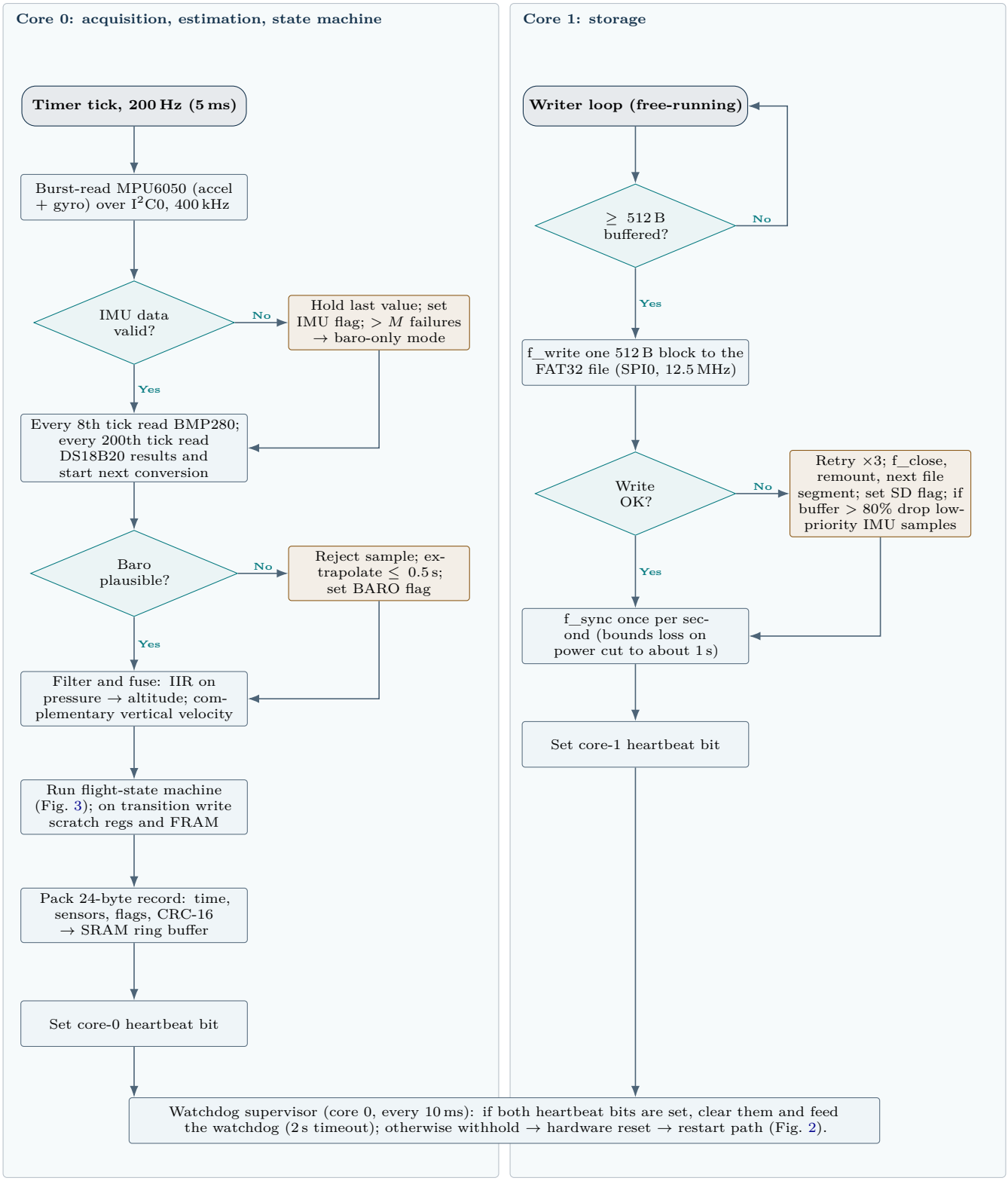


Figure 4: Logger Pico dual-core task flow with fault handling and the cross-core watchdog supervisor. Core 1 absorbs SD latency so core 0 timing is never blocked.

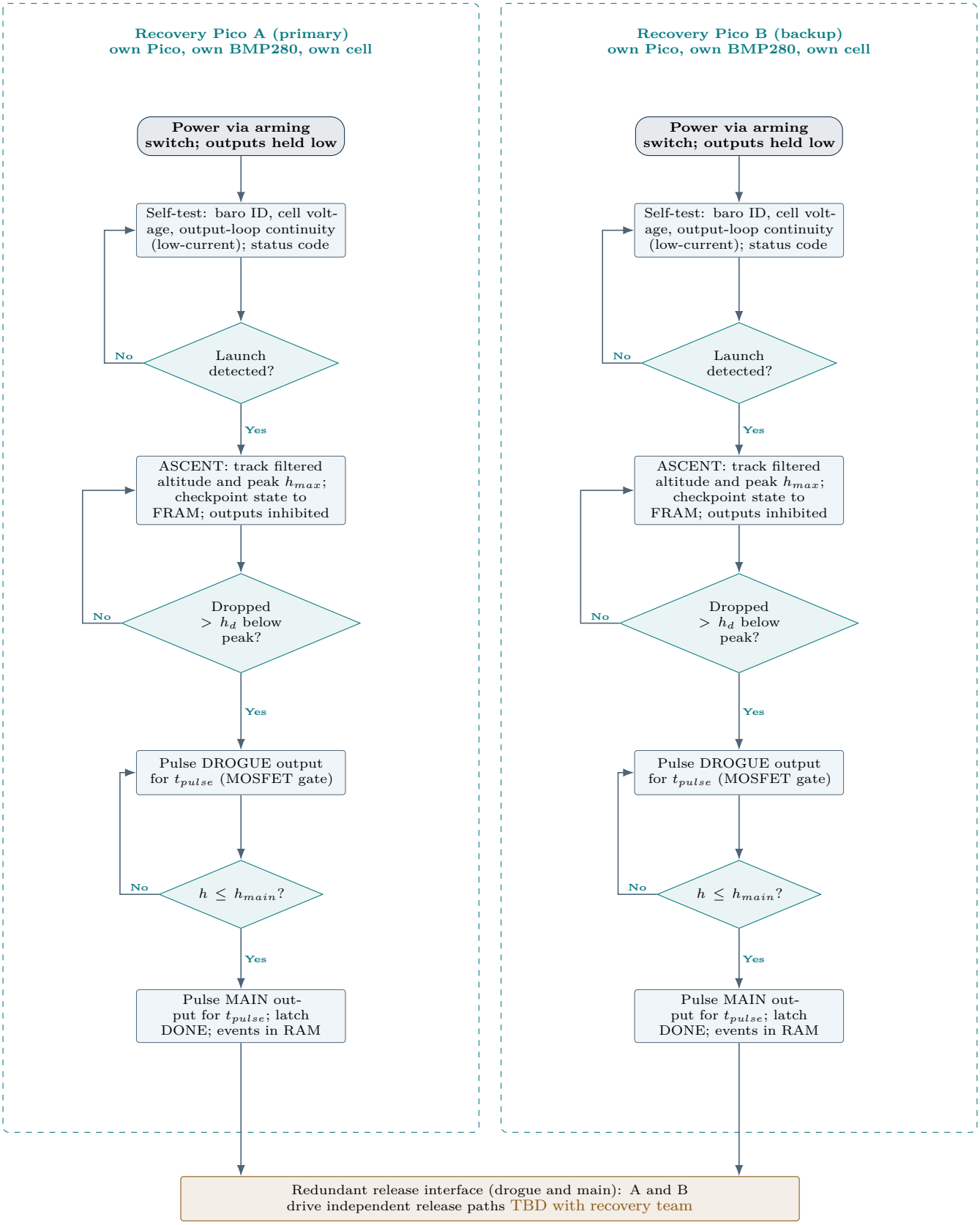


Figure 5: Independent recovery controllers. B uses a different barometer part, a later apogee trigger, a lower main altitude and an additional time-since-launch back-up compared with A (Secs. 7.1 and 8).

10. State preservation across resets and power interruption

A watchdog is not a power-loss protection. The RP2040 watchdog scratch registers persist through a *watchdog reset*, but they must not be assumed to survive a power-on reset, a loss of supply or a brown-out [6]; the exact behaviour for an external reset (RUN pin) should be re-read in the datasheet before the design is frozen. The design therefore treats the reset causes separately.

Event	Scratch registers	State-preservation strategy
Software lock-up caught by watchdog	Retained [6]	Restore from scratch registers only if magic number and CRC are valid; otherwise fall through to the FRAM checkpoint.
Brief supply dip or brown-out	Not relied on	Hold-up capacitance rides through short dips; if the Pico still restarts, restore from the FRAM checkpoint.
Complete power interruption (cell or connector)	Lost	Cold boot, restore from the FRAM checkpoint (if valid), set POWER_LOSS flag, open a new log segment.
External reset line pulse	Do not assume	Same as complete power interruption.
No valid checkpoint	Lost	Logger: restart as PAD or fallback per rule below. Recovery boards: see cold-boot rules.

Checkpoint record. A 16-byte record is written to an I^2C FRAM [20] on a bus (I^2C1) separate from the sensors: magic and version, flight state, launch-detect time, baseline pressure, peak altitude h_{max}, event flags, sequence counter and CRC-16. It is written at every state transition and at least every 250 ms, alternating between two slots so that a write cut short by power loss leaves the other slot intact. At 400 kHz a 16-byte write with 3 bytes of address overhead takes about $(16 + 3) \times 9/400\,000 \approx 0.43$ ms [13]. No flash is written in flight [6].

Hold-up. A bulk capacitor across the board supply keeps the rail up during a brief interruption: $C \approx I\,\Delta t/\Delta V$. For example, 60 mA for 20 ms with 0.5 V allowable droop needs about 2.4 mF. The value is to be sized and tested, not assumed.

Cold-boot rules for recovery boards (proposed).

- Outputs are held low from the first instant of boot by the hardware pull-downs; software never energises an output during start-up.
- With a valid checkpoint in an in-flight state, restore state and continue the normal logic; outputs stay inhibited until their trigger conditions are satisfied again.
- With no valid checkpoint, the board cannot know the pad baseline. A fallback rule is needed from the recovery lead; one candidate is to inhibit the drogue unless a sustained descent is measured, and to use the main-altitude trigger with the nominal site pressure. The other channel provides primary protection rule TBD.

Verification is by supply-interruption tests in every state (V-04).

11. Additional payload: thermal-environment mission definition

Research question. How does the temperature difference between the avionics bay and the inside surface of the airframe wall evolve through the pad, ascent, coast and descent phases of the 3K flight?

Hypothesis and null result. The difference $\Delta T(t) = T_{wall}(t) - T_{bay}(t)$ changes between phases by more than the measurement uncertainty. A flight in which no change larger than the uncertainty is observed is also a valid, reportable result.

New information. The data show whether the avionics bay stays well inside the Pico operating range of -20 to $+85$ °C [5], how large the bay-to-wall gradient is during flight, and how quickly the bay follows the structure. This supports placement and thermal margin decisions for later flights. The payload stays inside the rocket, is read by the logger and has no radio or stored-energy device; the in-body configuration with avionics communication throughout flight is the one proposed [1] (Sec. 7.1.1).

Item	Definition (team proposal)
Measurements	T_{bay} and T_{wall} from two DS18B20 probes [15], 1 Hz each, 12-bit, with sensor ID, timestamp, flight state and validity flag.
Phases	Pad (baseline), ascent (launch to burnout), coast (burnout to apogee), descent (apogee to landing), all from the logger state record (Fig. 3).

Item	Definition (continued)
Recording window	From arming through landing plus a post-landing period; expected duration from flight simulation, well inside the > 1.5 h battery endurance.
Accuracy	Each probe is specified at $\pm 0.5\,^{\circ}$C from -10 to $+85\,^{\circ}$C [15]; uncalibrated, ΔT is within $\pm 1.0\,^{\circ}$C worst case ($\approx 0.7\,^{\circ}$C root-sum-square). After both probes are offset-calibrated in a common stable bath, the target is a ΔT uncertainty of $\leq 0.3\,^{\circ}$C, to be demonstrated in V-08.
Response time	Probe step response measured on the bench so that lag between bay and wall probes is not mistaken for a physical gradient.
Success	Both channels valid for $\geq 90\%$ of expected samples in each of the ascent, coast and descent phases; records retrieved with CRC pass $\geq 99\%$; phase-mean ΔT and uncertainty reported.
Partial success	One channel valid $\geq 90\%$, or both channels valid for 50–89% of expected samples in at least two phases.
Failure	Fewer than 50% valid samples in the flight window, no retrieved file, or data that cannot be aligned to flight events.
Retrieval evidence	Recovered SD card, intact raw file with hash, processed $\Delta T(t)$ plot per phase, completeness table and quality log (Sec. 15).

12. Robustness and fault handling

Failure mode	Consequence	Mitigation (see flowcharts)
Logger lock-up	Gaps in data or payload record.	Watchdog restart with scratch-register restore; 1 s sync.
Power interruption in flight	Cold boot, loss of state and baseline.	FRAM checkpoint, hold-up capacitance, new log segment with POWER_LOSS flag (Sec. 10).
Barometer noise, spike or port blockage	False apogee or main trigger.	Plausibility gate, IIR filter, multi-condition apogee, lock-out, time-out.
IMU saturation	Bad launch/burnout and acceleration data.	Range decision gate (Sec. 9), saturation flag, baro cross-check.
SD write stall or removal	Missing records.	Core-1 offload, 24 kB buffer, retry/remount, next-file segment.
Recovery A failure	No or late deployment.	Backup B: separate board, diverse sensor, cell, later trigger and time back-up.
Common-cause failure of A and B	No deployment on both channels.	Diversity, release requirements R1–R4, fault-injection tests (Sec. 8).
Beacon track missing or misaligned	Incomplete flight-data package.	Redundant altitude sources, time-alignment check, quality flags (Sec. 15).
Premature output from noise/boot	Unsafe activation.	Outputs held low by pull-downs, arming switch, launch-inhibit logic.
Thermal probe detaches	Invalid payload data; loose object.	Mechanical retention, strain relief, flag in record.
Connector loosening	Intermittent power/data.	Locking connectors, strain relief, vibration test [1].

The full FMEA follows the organiser template [21]; classical FMECA practice is described in [22].

13. Power and data budgets

Load estimate (logger). Values are datasheet-typical or assumed, and must be replaced by measurements.

Item	mA	Basis
Pico L at 125 MHz, both cores	30	Assumed; measure [6]
MPU6050 (gyro + accel)	3.9	Datasheet typical [12]
BMP280	< 0.1	Datasheet [11]
2× DS18B20 active	2	About 1 mA each [15]
microSD module average	20	Assumed; write peaks higher
Total estimate	**≈56**	Design basis kept at **200 mA**

At 56 mA the 1.5 h minimum [1] needs about 84 mAh. Using the conservative 200 mA at 3.3 V through an assumed 85% efficient converter, the demand is $3.3 \times 0.2 \times 1.5/0.85 \approx 1.16$ Wh, against about 9.6 Wh stored in a nominal

3.7 V, 2.6 Ah cell (≈8× margin). The margin is only a sizing indicator; a measured endurance test of more than 1.5 h is required. Recovery boards use the same method plus the release-device pulse load TBD.

Data budget. 24 B × 200 Hz = 4.8 kB/s; 90 min = 5400 s ⇒ about 26 MB. SPI at 12.5 MHz and 512 B block writes leave large margin, and the 24 kB buffer covers about 5 s of SD latency.

14. Preliminary electronics budget and mass

Unit prices for sensors, boards and cells are carried from the earlier Shopee-linked bill of materials (listings recorded 4 November 2026, RM, before shipping, non-binding). Everything else is shown as a planning allowance or left unquoted. A realistic envelope must separate these three kinds of number.

Item	RM	Basis
Logger and thermal payload (10 listed parts 105.71 plus 60.00 allowances)	165.71	Shopee-priced parts plus allowances
Recovery A: Pico 15.27, BMP280 4.47, cell 29.99, board 4.49, connectors 4.40 (58.62) plus 65.00 allowances (protection and switch 25, mounts 20, MOSFET and sense parts 10, assembly 10)	123.62	Shopee-priced parts plus allowances
Recovery B: Pico 15.27, cell 29.99, board 4.49, connectors 4.40 (54.15) plus 80.00 allowances (MS5611 module 15, protection 25, mounts 20, MOSFET and sense 10, assembly 10)	134.15	MS5611 price not quoted; allowance
FRAM checkpoint memory, three boards at 10.00	30.00	Allowance, not quoted
Spares: Pico 15.27, BMP280 4.47, MPU6050 6.60, cell 29.99	56.33	Shopee-priced
Test and integration consumables (dummy loads, cables, fixtures)	40.00	Allowance
Subtotal	**549.81**	
25% contingency	137.45	Price variation, rework, spares
Planning envelope, excluding unquoted items	**687.26**	

Not included (unquoted): recovery release hardware and its initiators, the organiser-standard GPS beacon and antenna [1], the optional ADXL375 (needed only if Sec. 9 so decides), specialist qualification testing, and shipping. These must be quoted before the budget can be called complete.

Preliminary mass estimate. The masses below are assumed typical values and must be replaced by weighed values; the estimated uncertainty is ±30%.

Domain	g	Contents (assumed)
Logger and payload	119	Pico, BMP280, MPU6050, SD module and card, two probes with cable, cell, board, harness, mounts
Recovery A	96	Pico, barometer, cell, board, harness and drivers, mounts
Recovery B	96	As A
FRAM boards (three)	3	Small breakouts
Total electronics	**≈314**	Excludes release hardware, GPS beacon and long wiring runs

15. Flight-data interface and post-flight evidence package

The organiser beacon is physically independent of the Pico boards, but the four required outputs (altitude, velocity, acceleration and GPS trajectory [1], Sec. 2.1) must end up as one usable evidence package. The interface is therefore defined at the data level.

Output	Primary source	Cross-check	Availability criterion
Altitude	Logger barometer, filtered	Recovery A and B barometers; beacon altitude if provided	Valid for ≥ 90% of the flight window

Output	Primary source	Cross-check	Criterion (continued)
Velocity	Derivative of filtered barometric altitude	Integrated acceleration; beacon-derived speed if provided	Valid for $\geq$ 90% of the flight window
Acceleration	MPU6050 (and ADXL375 if fitted)	Second derivative of altitude (coarse)	$\geq$ 90% valid, saturation flagged
GPS trajectory	Organiser beacon track file	Barometric altitude trend	Track covers launch to landing; retrieval method to confirm with the organiser

Time alignment. The logger stamps every record with its own microsecond timer and records the launch-detect event T_0. The beacon track carries UTC. The two are aligned at T_0 and refined by correlating barometric altitude with beacon altitude during ascent. A crystal tolerance assumed to be 50 ppm would give up to $5400 \times 50 \times 10^{-6} \approx 0.27$ s drift over 90 min, which is small compared with the beacon update interval. An operator-noted UTC at ignition (to about 1 s) is kept as an independent check. The achieved residual is reported with the data.

16. Verification plan

Methods follow the usual inspection, analysis, demonstration and test categories [4]; MDR needs the plan, not the results [3].

ID	Method	Planned evidence	Initial pass criterion (team target)
V-01	Test	Replay recorded and simulated histories through the A and B logic with injected noise and spikes.	No false apogee or main trigger in $\geq$ 1000 randomised replays.
V-02	Test	Logger timing: loop jitter, buffer occupancy, SD latency, forced SD errors.	Jitter $<$ 10% of the 5 ms period; no overruns; buffer $<$ 50% full.
V-03	Demonstration	Force a core stall mid-sequence; confirm watchdog restart and state restore from scratch registers.	State correct; log gap $\leq$ 3 s.
V-04	Test	Interrupt supply for 1 ms, 20 ms, 100 ms and 1 s in every flight state on logger and recovery boards.	Correct state restored from FRAM; outputs stay low throughout; POWER_LOSS flagged.
V-05	Demonstration	Fault injection with safe test loads: A disabled, B disabled, both enabled.	Each channel alone releases drogue then main in 10 of 10 trials.
V-06	Test	Power endurance with realistic logging; connector pull and vibration retention.	Runs $\geq$ 1.5 h with $\geq$ 25% capacity margin.
V-07	Analysis	Acceleration envelope against thrust curve, mass and deployment shock (Sec. 9).	$k\,a_{pk} \leq$ 16 g, or ADXL375 fitted.
V-08	Test	Thermal probes: common-bath offset calibration, step response, 1 h data-completeness run.	ΔT uncertainty $\leq$ 0.3 °C; completeness $\geq$ 99%.
V-09	Test	Barometer ports: pressure chamber or vented-bay test with representative pressure profile.	A and B altitude agree within $\pm$5 m.
V-10	Demonstration	Dry run of the post-flight pipeline (Fig. 6) on a ground or simulated data set.	All four outputs produced; alignment residual $\leq$ 1 s.
V-11	Inspection	Wiring, labels, strain relief, arming-switch access and polarity [1].	No findings.
V-12	Analysis	Mass, power and cost budget and FMEA update for PDR [21].	Budget and FMEA approved by lead.

17. Decisions needed from the team

1. Select the release-interface arrangement (R1–R4) with the recovery lead, and approve the diverse channel B (MS5611, separate firmware, time back-up).
2. Provide the Pioneer-10K thrust curve, mass history and flight simulation so that a_{pk} and the ADXL375 decision (Sec. 9) can be made, and so that a_L, t_{lock}, v_{max} and $T_{apo,max}$ can be fixed.
3. Approve the FRAM checkpoint and agree the recovery-board fallback rule when no valid checkpoint exists.

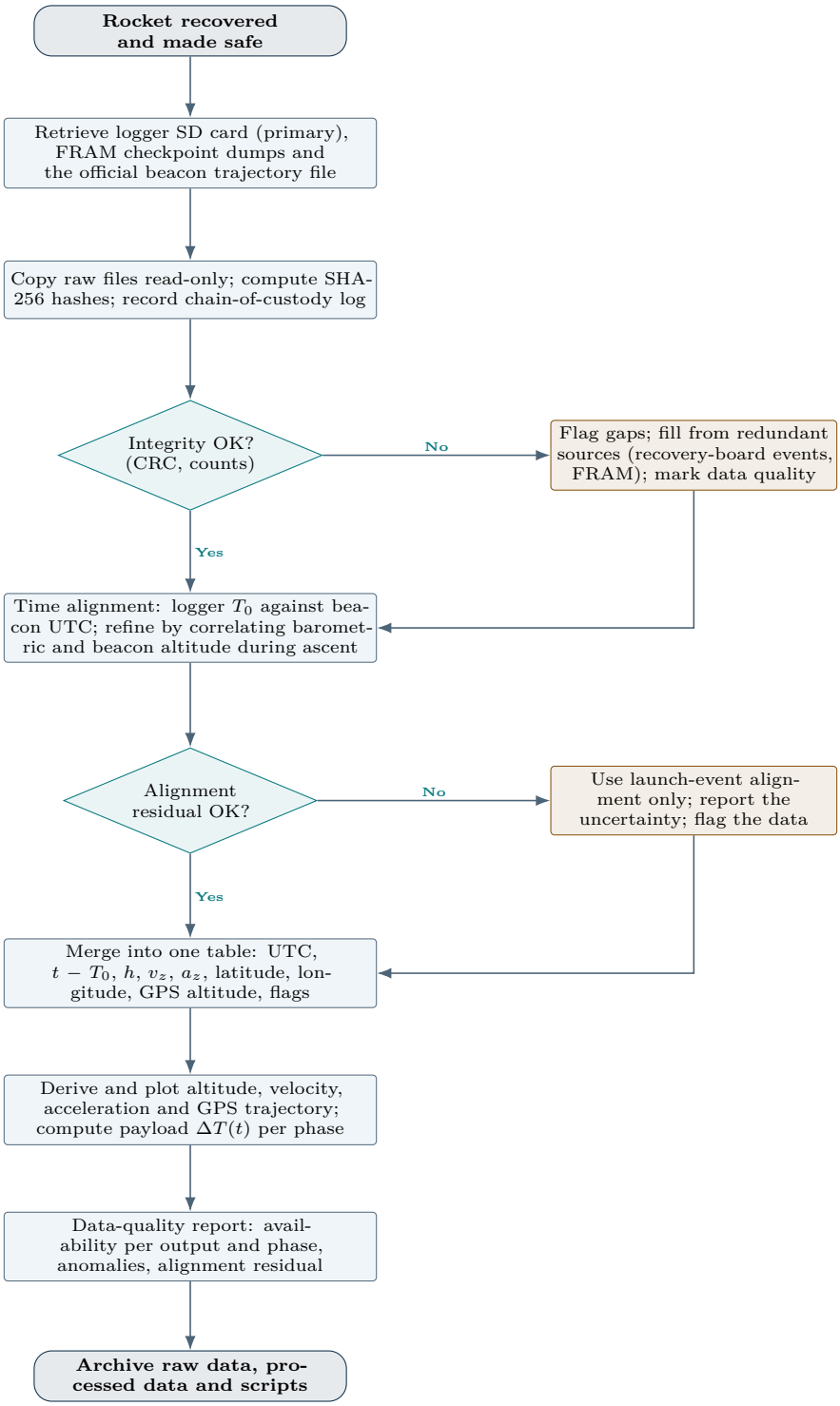


Figure 6: Post-flight data pipeline that turns the logger files and the organiser beacon track into one evidence package covering all four required outputs.

4. Approve the thermal research question and success criteria, or replace them with an agreed payload using the same format.
5. Obtain quotes for release hardware and the GPS beacon, and confirm how the beacon track file is retrieved.
6. Confirm bay dimensions, mounting, battery protection and measured current before procurement.

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